How Does Family Involvement Influence Student Academic Performance?

STA-631: Classification Project

18 April 2026

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Data source	2019 Parent and Family Involvement (PFI) survey, curated 2019 sheet

Bottom line: home-based engagement appears more consistently related to stronger academic performance than school-based participation alone. The lasso logistic model was the strongest binary classifier, and the multinomial model was the best way to preserve the original four-level grade outcome.

Executive Summary

This report explains the purpose of the work, the data decisions, the exploratory findings, the modeling workflow, the strengths and limitations of each model, and the practical recommendations that follow from the analysis.

- Research question: how are school-based and home-based forms of family involvement associated with student academic performance?
- Data choice: the 2019 PFI sheet was used because it was the latest and offered the richest feature set for this problem.
- Main findings: homework-related and home-support variables showed a clearer relationship with academic success than broad school-event participation variables.
- Best binary classifier: lasso logistic regression (accuracy 0.715, ROC AUC 0.773).
- Best full-outcome model: multinomial regression (accuracy 0.624, multiclass ROC AUC 0.712).
- Poisson regression was demonstrated on FSFREQ, but the variance-to-mean ratio of 12.83 indicated serious overdispersion, so the report recommends a negative binomial model as the better next-step count model.
- The dominance of the Mostly A and Mostly B categories means accuracy must be interpreted cautiously; ROC AUC, confusion patterns, and calibration-related discussion are more informative than raw accuracy alone.
- The project is predictive and associative, not causal. It identifies patterns that can guide advice, not proof that one family behavior directly causes better grades.
- Practical implication: K-12 support efforts should prioritize repeatable home learning routines, not only school-event attendance.

Project Purpose and Why this Study Matters

The project was framed around a real advising problem for GVSU's K-12 Connect: identify which patterns of family involvement appear most closely tied to student performance, then use those patterns to make clearer recommendations for schools and families. The original analysis focused on parent-reported grades as the main academic outcome and contrasted school-based involvement with home-based engagement.

That framing matters because “family involvement” is not one behavior. Some family actions happen in the school building, such as attending activities or contacting staff. Other actions happen in the home, such as maintaining homework routines, checking

work, or creating everyday learning support. The report therefore treats the project as both a predictive modeling exercise and a practical interpretation exercise.

Data Source, Variables, and Analytic Framing

The analysis used the curated 2019 Parent and Family Involvement (PFI) survey file. In the uploaded workflow, the starting dataset contained 15,500 rows and 75 variables. After cleaning, recoding, and filtering for required predictors, the main modeling file used for the core analysis contained 13,533 rows and 42 variables.

Analytic fact	Value
Starting file size	15,500 rows, 75 variables
Main modeling file	13,533 rows, 42 variables
Outcome missingness	SEGRADES missing in 12.7% of records
Homework-variable missingness	About 6.8% for several homework variables
Outcome versions	Binary high-achiever indicator and four-level grade outcome

The central variables came from four groups: academic outcomes (especially SEGRADES), school-based involvement items, school-contact items, and home-engagement variables such as homework frequency, homework time, homework checking, and family dinner frequency. Demographic and household controls such as parental education, income, sex, disability status, age, household size, and race/ethnicity were also carried into the models.

What We Did to Load, Clean, and Prepare the Data

- Imported the curated 2019 Excel sheet and inspected the dimensions, variable names, and variable structure.
- Converted negative numeric codes to missing values so nonresponse and inapplicable values did not behave like valid numbers.
- We used variables describing parent-reported grades, school-based family involvement, school contact, homework and home engagement, school choice, and household background. Key examples included SEGRADES, FSSPORTX, FSVOL, FSMTNG, FSNOTESX, FSMEMO, FHHOME, FHWKHRS, FODINNERX, PARGRADEX, TTLHHINC, RACEETH, CSEX, and DSBLTY.
- Treated the invalid SEGRADES code as missing, then built two response variables: a four-level grade outcome and a binary high-achiever indicator for students with mostly A's.
- Created composite scores for school involvement and school contact so related survey items could be summarized instead of interpreted one by one.
- Used complete-case analytic files within each modeling task to keep the workflow transparent across multiple model families.
- The complete-case approach also improved workflow transparency. Separate analytic files were created for binary classification, multinomial classification, and count modeling, with each file retaining only the variables necessary for that task. The tradeoff is that complete-case analysis can introduce bias if missingness is systematic rather than random.

This approach kept the preprocessing aligned with the modeling goals. The binary outcome supported direct comparison across lasso logistic, standard logistic, LDA, and QDA. The multinomial model preserved the full grade structure. Poisson regression was demonstrated on FSFREQ because it is a true count variable: the number of times adults attended school meetings or activities.

The missingness screen showed that the main response variable SEGRADES had the largest share of missing values at 12.7%. Homework-detail variables such as FHWKHRS, FHAMOUNT, and FHCAMT had smaller but still noticeable missingness at 6.8%,

while many school-involvement variables had less than 1% missingness. That pattern matters because it means the data reduction was driven more by the outcome and selected homework variables than by the broad school involvement block.

Variable	n missing	% missing
SEGRADES	1,967	12.7
grade_multi	1,967	12.7
FHWKHRS	1,047	6.8
FHAMOUNT	1,047	6.8
FHCAMT	1,047	6.8

From a statistical standpoint, this preprocessing step is also where several modeling assumptions are implicitly shaped. Cleaning invalid codes, choosing factor reference levels, and deciding whether to collapse categories all affect the eventual interpretation of coefficients, margins, and predictive metrics.

Exploratory Data Analysis and What the Visuals Revealed

The exploration stage established the main story before formal modeling began. Three patterns stood out: the grade outcome was highly imbalanced, school-based involvement did not show a simple positive relationship with stronger grades, and homework-related engagement looked more promising as a practical predictor.

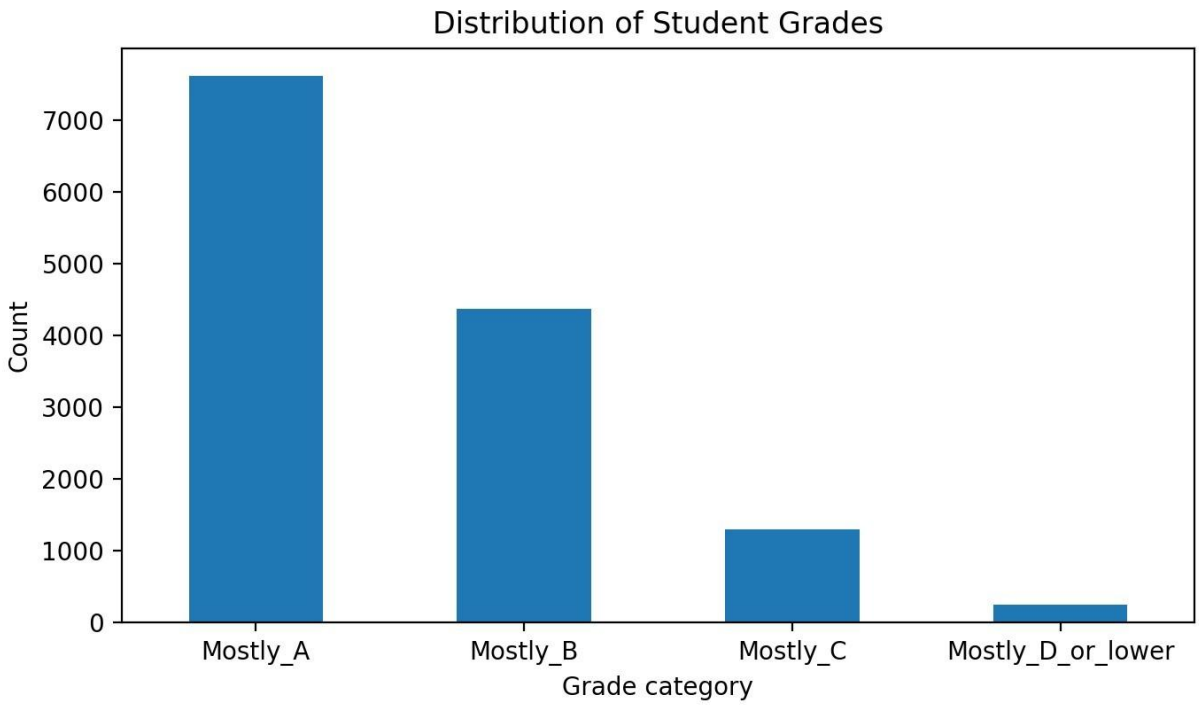


Figure 1. Distribution of student grades in the cleaned analytic file.

This bar chart shows the outcome imbalance clearly. Mostly A and Mostly B dominate the response, while Mostly C and especially Mostly D-or-lower are much less common. That is statistically important because a classifier can look respectable on raw accuracy

by learning the majority categories well, even when it performs poorly on the rarer and often more policy-relevant lower-performing groups. This is one reason the report compares models using ROC AUC and confusion behavior in addition to accuracy.

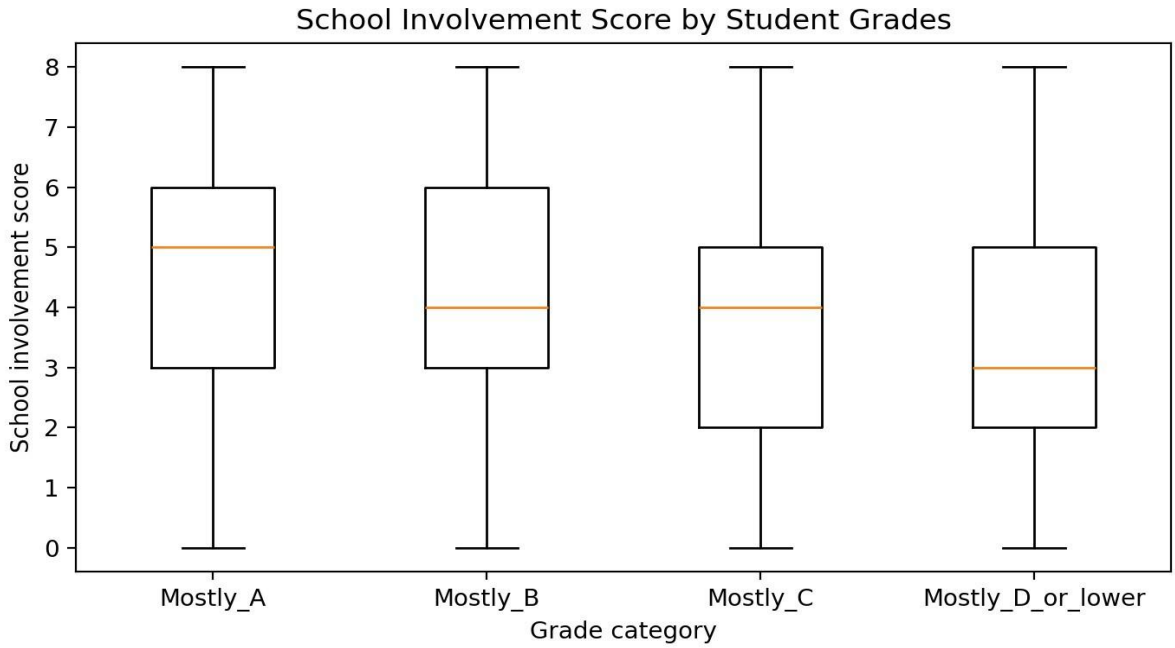


Figure 2. School involvement score by grade category.

The boxplot for school_involve_score does not display a clean monotone increase from lower grades to higher grades. In fact, the median school-involvement score is similar across groups and can even be slightly higher for lower-performing students. This weakens any simple claim that attendance at school events is independently predictive of stronger grades. A more plausible interpretation is reactive involvement: adults may become more engaged with school when students are already experiencing academic difficulty.

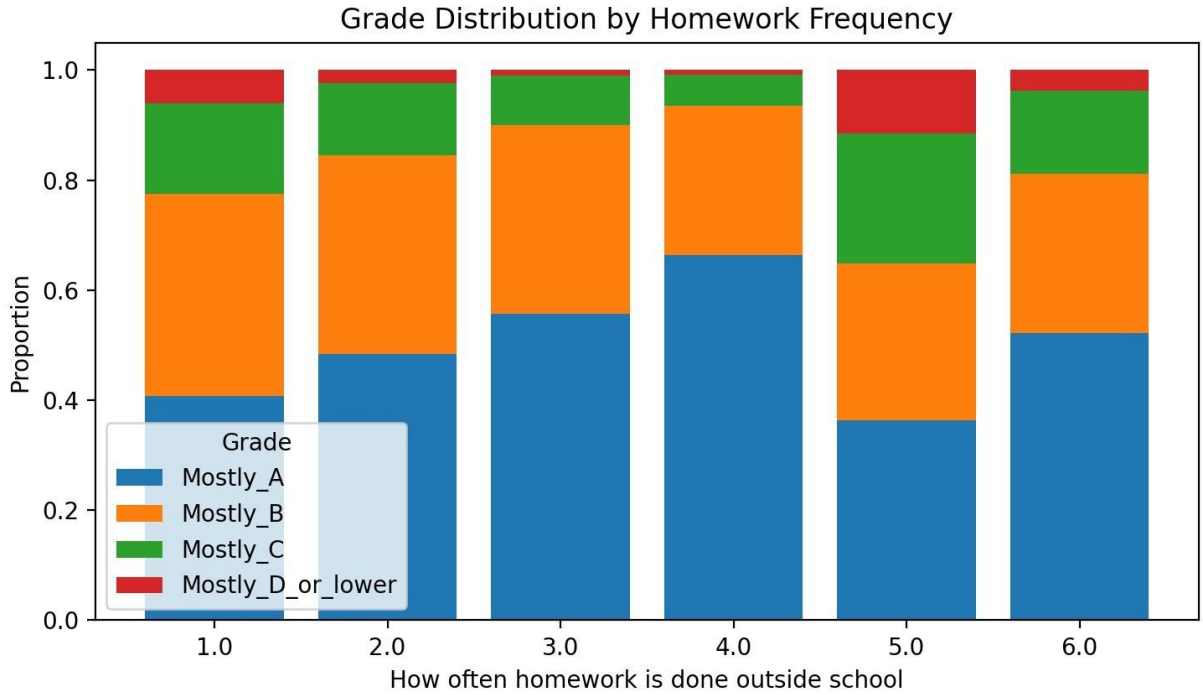
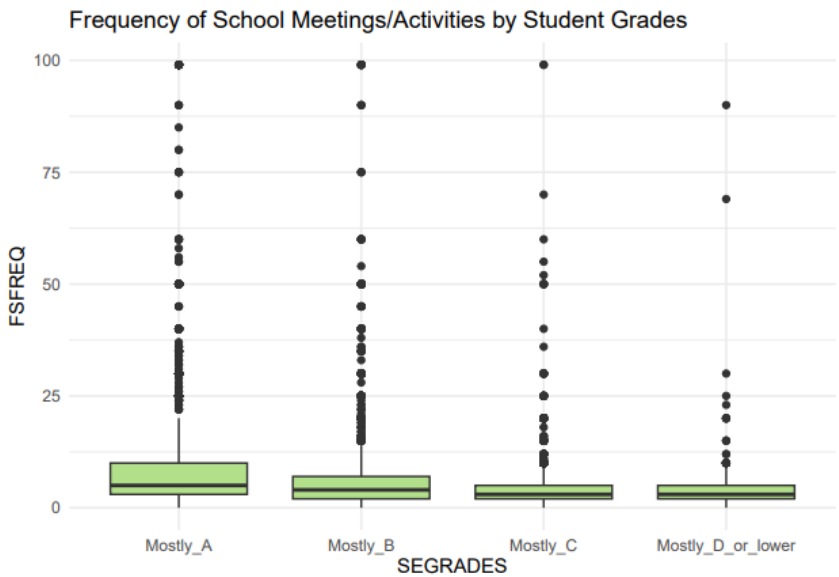


Figure 3. Grade distribution by homework frequency.

Homework frequency shows a clearer relationship with academic performance. Higher levels of homework completion outside school correspond to a larger share of Mostly A students, which is one reason the home-engagement variables became important in later model interpretation.



The frequency of school meetings and activities shows a weak positive association with academic performance, but the large variability and presence of extreme values suggest that this relationship is not strong and that the distribution is highly skewed.

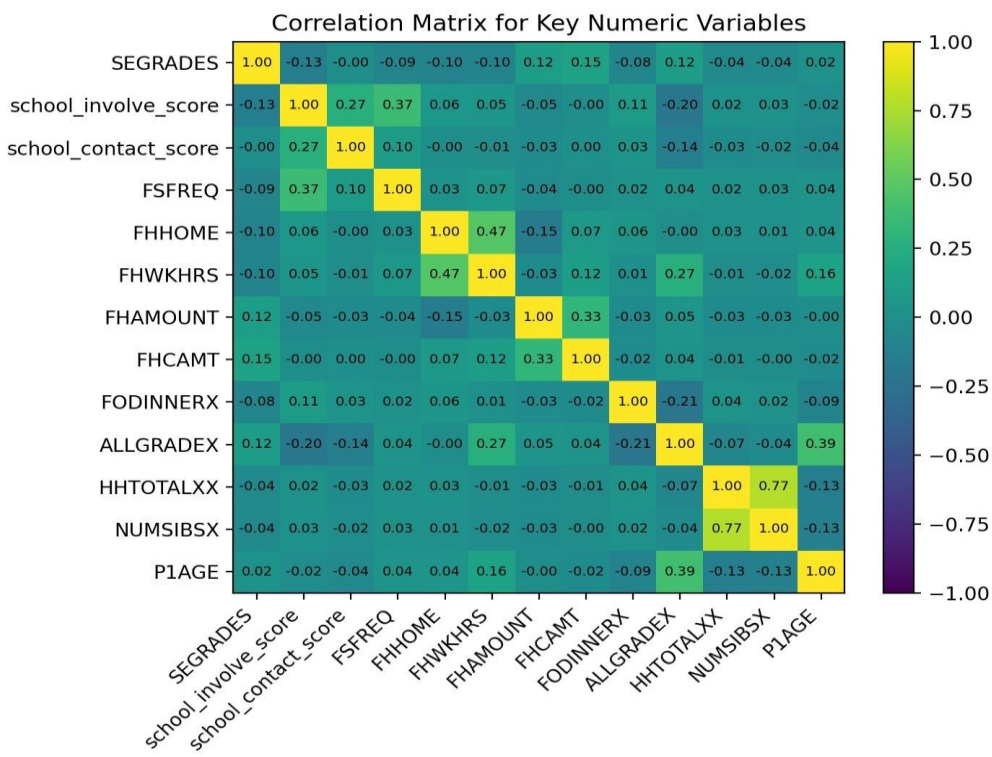


Figure 4. Correlation matrix for key numeric variables used in the analytic workflow.

There is no single overwhelming predictor that explains grades by itself. Instead, the relationships are mostly small to moderate in magnitude. That is important because it suggests this is a multivariable prediction problem rather than a one-variable story. Several moderate relationships do stand out: FHHOME and FHWKHRS are positively related to each other, and HHTOTALXX and NUMSIBSX are strongly related, which warns us that household-size variables may carry overlapping information. The relatively modest pairwise correlations also help explain why regularized logistic regression is attractive: it can stabilize estimation in the presence of multiple weak-to-moderate predictors without requiring that any one variable dominate.

Interpretation of the Main Predictors

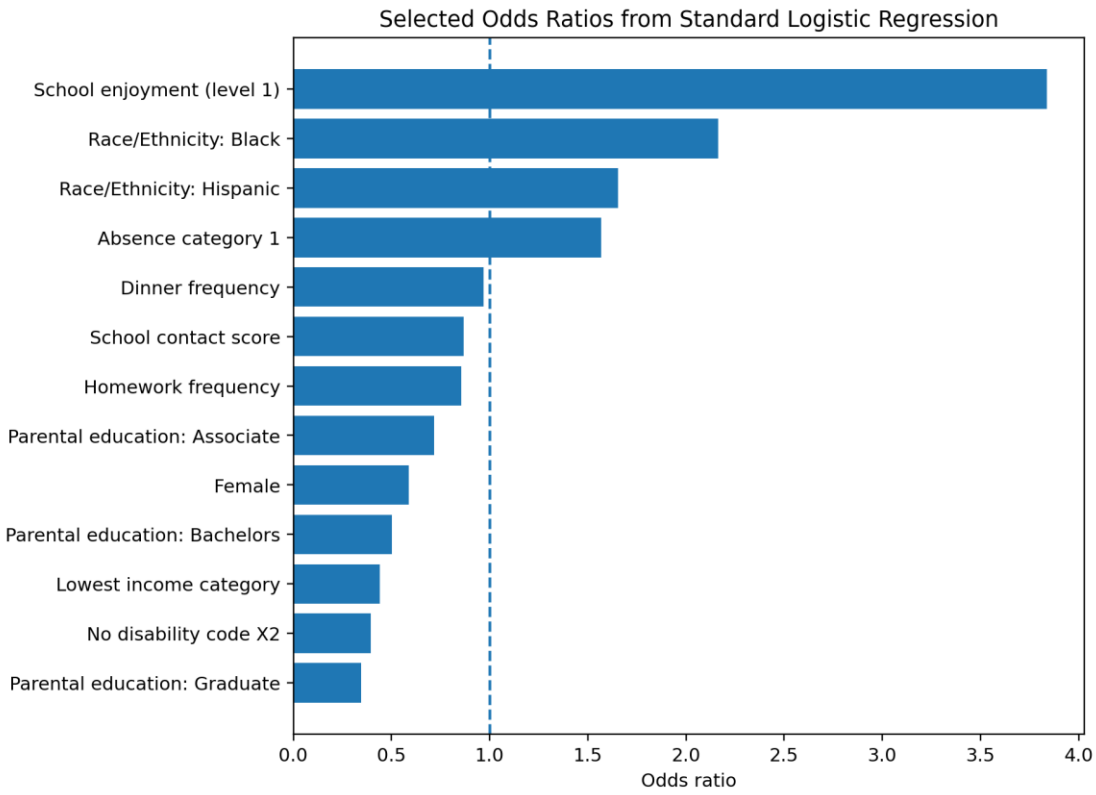


Figure 3. Selected odds ratios from the standard logistic regression model. Reference-category coding in the original model should be kept in mind when interpreting direction and magnitude.

The evidence points to a distinction between everyday learning support at home and formal involvement with school events or contacts.

- Home-based engagement shows the clearest and most consistent connection to stronger grades.
- School-based contact remains relevant, but it appears weaker and less stable as a predictor.
- School-based involvement may be reactive, increasing when academic concerns are already present.
- Student experience variables, especially school enjoyment and absenteeism, are also highly influential.
- Household context matters as well, including parental education, income, disability status, sex, and race or ethnicity.

Interpretive takeaway: The strongest evidence in this project supports regular day-to-day engagement with learning at home as the more dependable signal of stronger academic performance.

Modeling Strategy

The project used several model families because each answered a different version of the same substantive question. Logistic and lasso logistic regression asked who is likely to be a high achiever. Multinomial regression preserved the original four-category grade outcome. LDA and QDA served as benchmark classification methods. Poisson regression demonstrated count modeling on school-meeting attendance frequency.

- Lasso logistic regression was tuned with 10-fold cross-validation over a penalty grid, then chosen by ROC AUC. This serves two goals at once: shrinkage for generalization and variable selection for interpretability.
- Standard logistic regression kept the classical generalized linear model structure so odds ratios could be interpreted directly.
- Multinomial regression retained the ordered academic categories as four nominal outcome levels, which is closer to the survey response than converting everything to yes/no.
- LDA and QDA provided benchmark classifiers with linear and quadratic decision boundaries, making it possible to see whether a different geometric assumption improved predictive discrimination.
- Poisson regression was used for FSFREQ because the outcome is a count. That was a defensible starting point, but the validity of that choice depends on the Poisson mean-variance assumption.

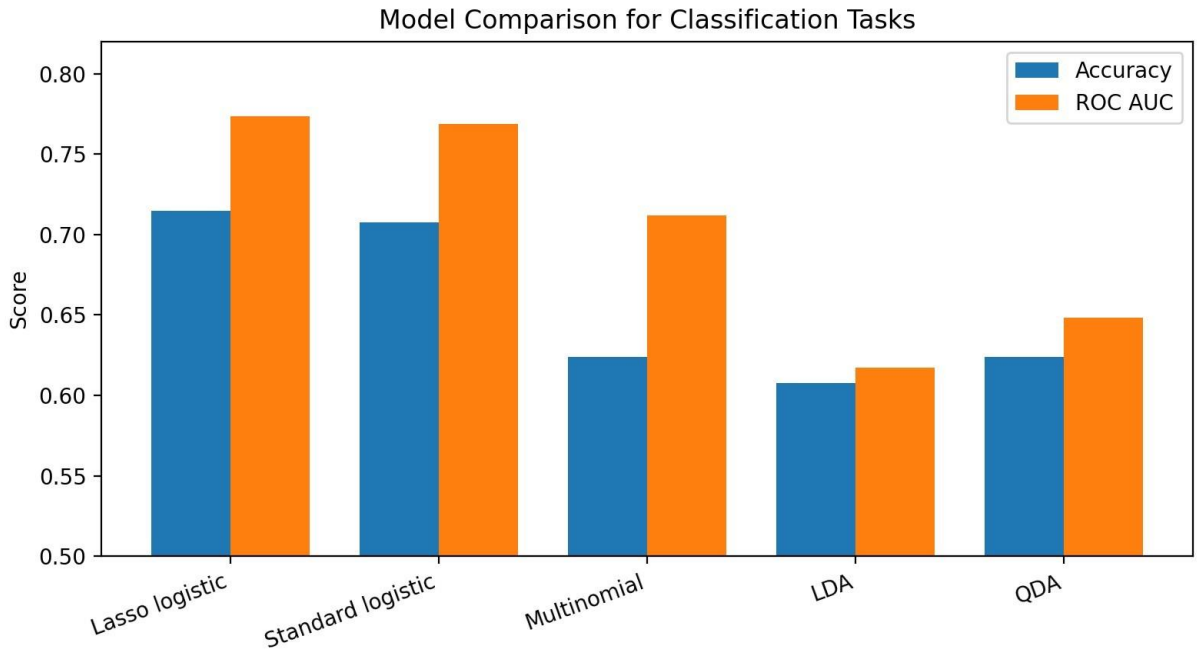


Figure 5. Accuracy and ROC AUC across the main classification models.

The grouped bar chart makes the ranking easy to see. Lasso and standard logistic are nearly tied at the top, with lasso holding a small but consistent edge. Multinomial regression is naturally lower because predicting four classes is harder than predicting two. LDA and QDA achieve weaker overall discrimination. The chart also communicates why model choice should depend on the task: the best binary model and the best full-outcome model are not the same object.

Model-by-Model Findings, Strengths, and Limitations

Model	Outcome	Accuracy / RMSE	ROC AUC	Strength	Main limitation
Lasso logistic	High achiever	0.715	0.773	Best binary performance and more parsimonious predictor set.	Still reduces grades to a binary outcome and does not imply causation.
Standard logistic	High achiever	0.708	0.769	Most interpretable odds-ratio model.	Slightly weaker than lasso and still limited to binary framing.
Multinomial regression	Four-level grades	0.624	0.712	Preserves the original grade categories.	Struggles more on rare lower-grade classes because of imbalance.
LDA	High achiever	0.607	0.617	Very high recall for identifying positives.	Weak discrimination and many false positives.
QDA	High achiever	0.624	0.648	Better than LDA on accuracy and AUC.	Still underperforms logistic based models.
Poisson regression	FSFREQ count	8.53 RMSE	—	Demonstrates the count modeling requirement with a substantively relevant outcome.	Severe overdispersion makes simple Poisson fit poor.

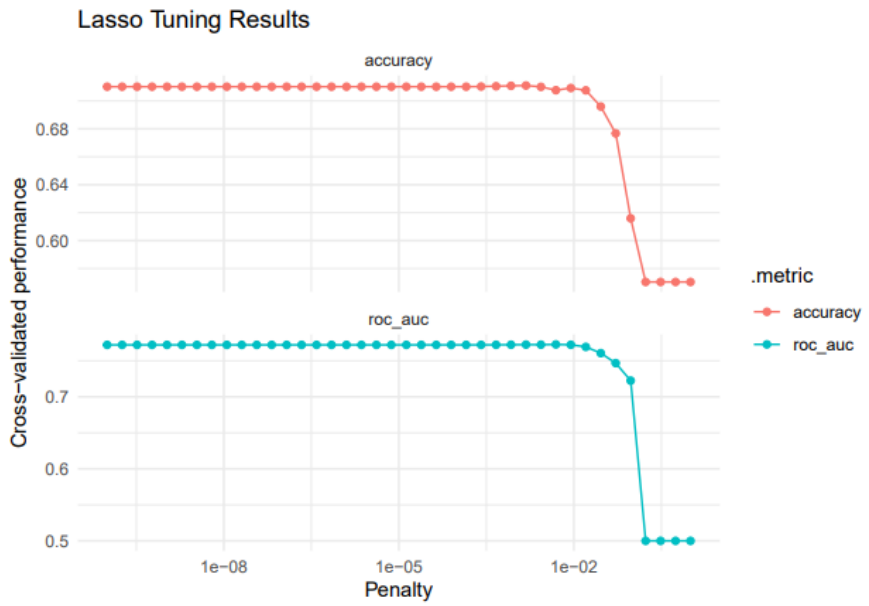
Lasso Logistic Regression

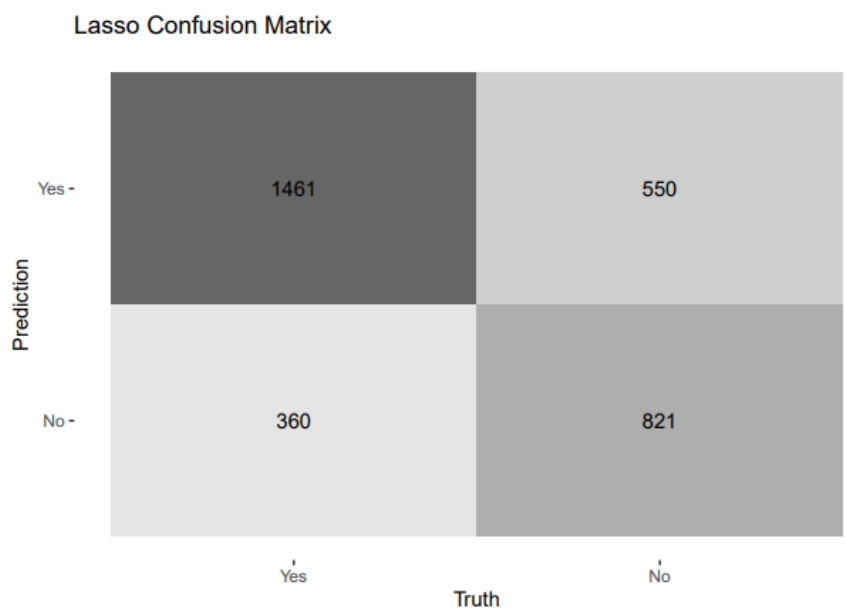
The lasso logistic model achieved the strongest binary-classification performance, with test accuracy of 71.5% and ROC AUC of 0.773. Its key advantage was not only prediction but parsimony:
it shrank weaker predictors to zero and highlighted a smaller set of variables that appeared to carry the most information.

Table 2: Top Lasso-Selected Predictors

term	estimate	penalty
SEENJOY_1	0.4124879	0.0049239
DSBLTY_X2	-0.3599257	0.0049239
PARGRADEX_Graduate	-0.3021157	0.0049239
CSEX_Female	-0.2390815	0.0049239
TTLHHINC_01	-0.2087388	0.0049239
RACEETH_Black	0.1939437	0.0049239
RACEETH_Hispanic	0.1761597	0.0049239
SEENJOY_2	-0.1480184	0.0049239
FHHOME	-0.1448516	0.0049239
PARGRADEX_Bachelors	-0.1344068	0.0049239
FHCAMT	0.1291649	0.0049239
SEABSNT_1	0.0760875	0.0049239
HHPARN19_BRD_X2	0.0718702	0.0049239
P1AGE	0.0649976	0.0049239
RACEETH_Asian_PI	-0.0640073	0.0049239
CENREG_X4	0.0613216	0.0049239
PARGRADEX_HS	0.0566932	0.0049239
CENREG_X2	-0.0492400	0.0049239
school_contact_score	-0.0451594	0.0049239
ZIPLOCL_X32	-0.0356906	0.0049239

The lasso model selected a relatively small set of predictors, indicating that only a subset of the available family, school, and demographic variables contributed meaningful predictive information. Among the retained predictors, student school experience variables such as school enjoyment and absenteeism, along with household background characteristics such as parental education, income, disability status, and demographic factors, showed the strongest effects. Importantly for the research question, home-based involvement variables such as homework-related engagement (FHHOME, FHCAMT) remained in the model, while school contact was 12 retained with a smaller coefficient. This suggests that family involvement at home may be more strongly associated with academic performance than school-based involvement, although broader student and household characteristics also play a major role. race/ethnicity remained in the selected model, indicating differences in predicted academic performance across groups after accounting for other retained covariates.

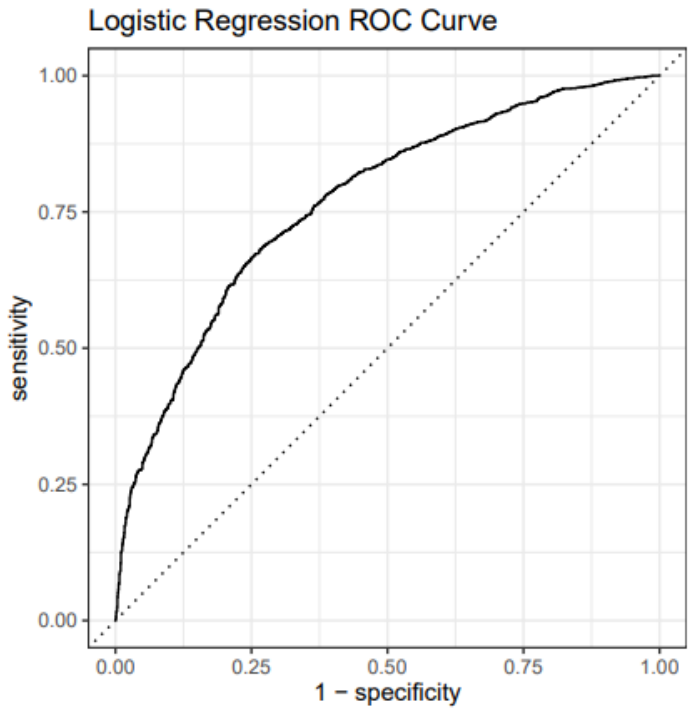




Standard Logistic Regression

The standard logistic regression model performed almost as well as lasso, with 70.8% accuracy, 0.769 ROC AUC, and a Brier score of 0.193. Its main strength is interpretability because odds ratios make it easier to explain which predictors increase or decrease the odds of being a high achiever, holding the rest constant.

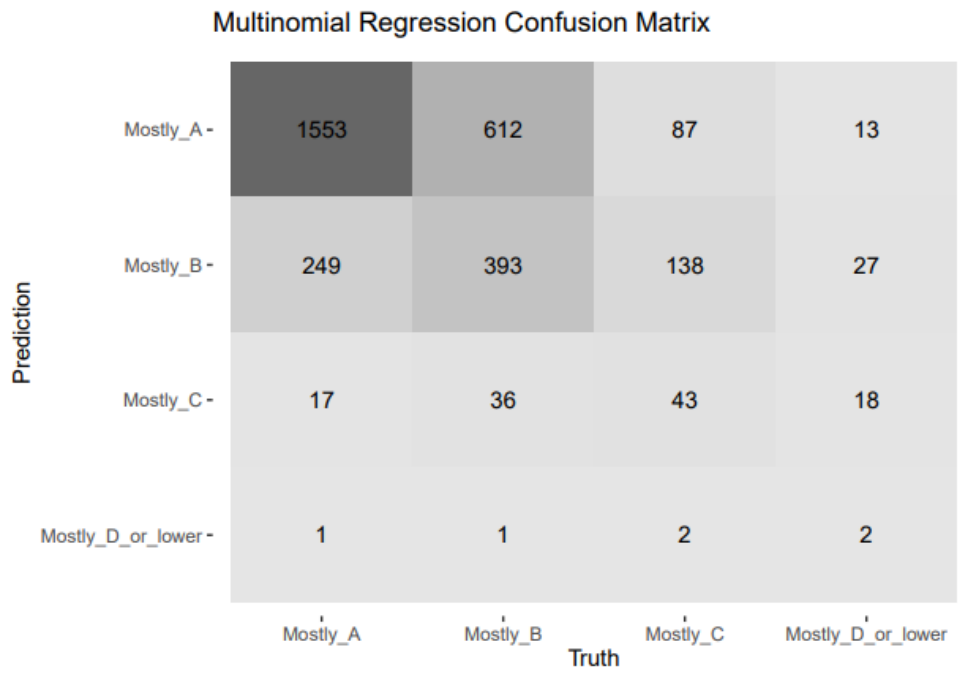
In the uploaded output, school contact score had a negative odds ratio, FHHOME also indicated lower odds as coded in the model, and student enjoyment of school showed especially large effects. Those details should be interpreted carefully because factor coding and reference categories matter, but the broader pattern is stable: home-support variables and student-experience variables are more informative than simple school-attendance measures.



Multinomial Regression

The multinomial model is the preferred model for communicating the original SEGRADES response because it does not collapse the outcome into a binary indicator. On the held-out test set it achieved accuracy of 0.624, multiclass ROC AUC 0.712, and Brier score 0.239. Those numbers are lower than the binary models, but this is expected because the task is much harder: the model must separate four categories rather than two.

Its confusion matrix is especially revealing. The model classified Mostly A very well and did a fair job on Mostly B, but it struggled much more on Mostly C and Mostly D-or-lower. That is exactly what the class distribution would predict: rarer categories are harder to learn because the model has fewer examples from which to estimate their class boundaries. This is not automatically a sign of overfitting. It is more directly a sign of class imbalance and overlapping among the middle-to-lower achievement groups.



Multinomial confusion matrix (prediction rows, truth columns)

Prediction	Mostly A	Mostly B	Mostly C	Mostly D-orlower
Mostly A	1553	612	87	13
Mostly B	249	393	138	27
Mostly C	17	36	43	18
Mostly D-or-lower	1	1	2	2

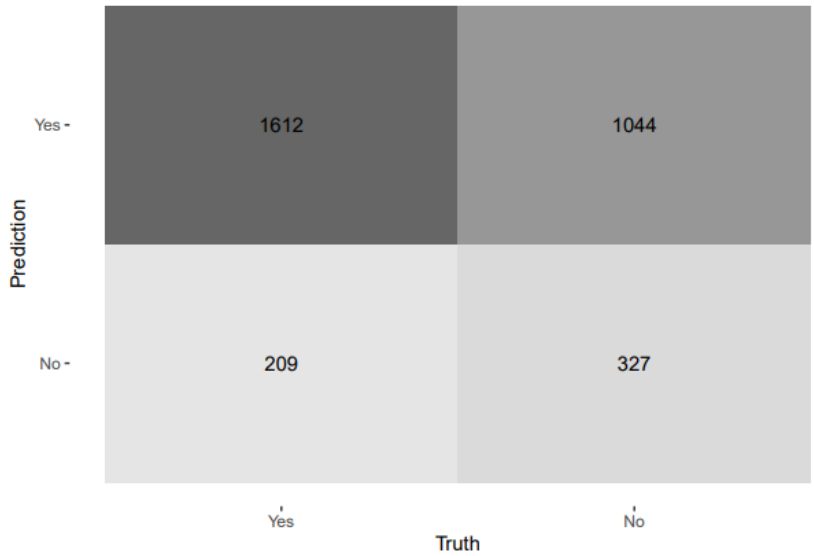
The matrix shows heavy concentration on the Mostly A and Mostly B rows, with only a small number of correct assignments in the rarest categories. The practical implication is that the multinomial model is useful for preserving the full response, but stakeholders should not read it as equally reliable across every grade category.

LDA and QDA

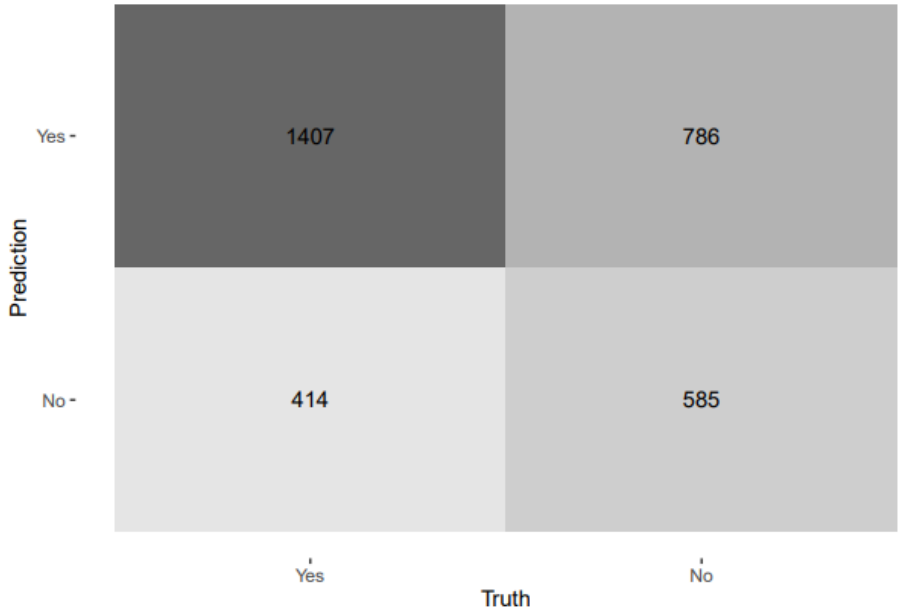
LDA and QDA served as useful benchmark classifiers rather than preferred final models. LDA achieved accuracy 0.607, precision 0.607, recall 0.885, and ROC AUC 0.617. QDA improved to accuracy 0.624, precision 0.642, recall 0.773, and ROC AUC 0.648. The striking feature is that LDA had very high recall but poor overall discrimination, which means it was relatively aggressive about labeling cases as high achievers. QDA softened that tradeoff somewhat, but both methods still trailed the logistic models.

In confusion-matrix terms, the discriminant models placed too much weight on catching positives, especially in the LDA case. That raised recall while also generating more false positives. For communication to GVSU K-12 Connect, that matters because a model that over-calls high achievers may be less useful for prioritization or intervention planning.

Confusion Matrix for LDA



Confusion Matrix for QDA



Poisson Regression

Poisson regression was fit to FSFREQ, the number of times adults in the household attended meetings or activities at the child's school. Substantively, that was a sensible count outcome. Statistically, however, the model did not satisfy a key assumption. The mean of FSFREQ was 7.19 and the variance was 92.28, yielding a variance-to-mean ratio of 12.83.

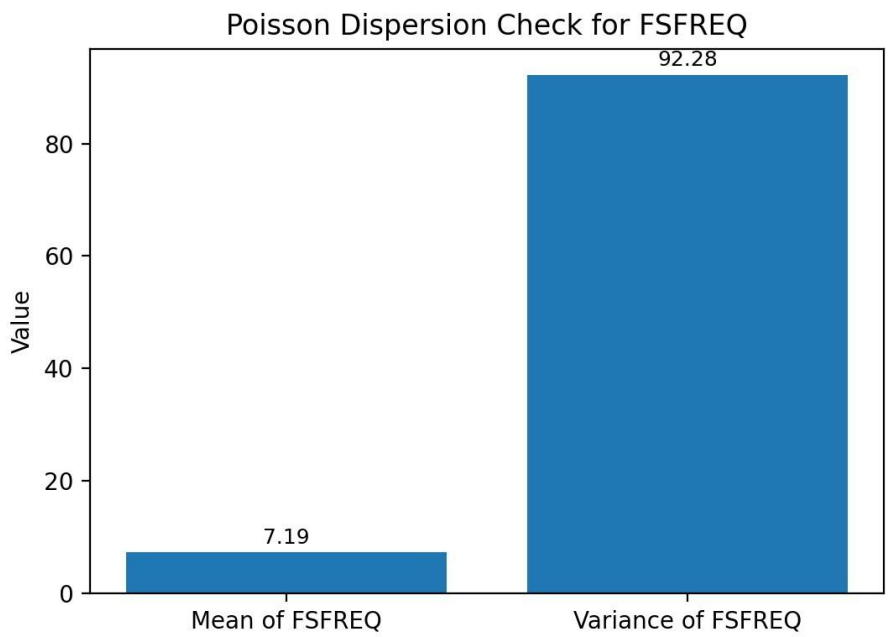


Figure 6. Mean-variance comparison for FSFREQ, showing strong overdispersion.

The dispersion check showed a mean of 7.19 and a variance of 92.28, for a variance-to-mean ratio of 12.83. Because Poisson regression assumes that the mean and variance are roughly equal, this is a major assumption failure. The test-set RMSE of 8.53 was also large relative to the mean, so the model was demonstrated but not retained as a preferred final model.

Table 4: Poisson Dispersion Check for FSFREQ

mean_fsfreq	var_fsfreq	ratio
7.190098	92.28142	12.83451

Overdispersion, Negative Binomial Thinking, and Why It Matters

Overdispersion is one of the most important modeling issues in the project because it changes the interpretation of the count-modeling component. A Poisson generalized linear model assumes E(Y| X) = Var(Y|X). Here, the empirical variance of FSFREQ is more than twelve times the mean. That is not a minor deviation; it is a major diagnostic warning.

In applied terms, overdispersion usually means one or more of the following: the data contains unmodeled heterogeneity, there are excess zeros or occasional very large counts, the event process is more clustered than Poisson allows, or important predictors are missing. Any of those mechanisms could plausibly occur for school-meeting attendance. Some families may rarely attend, some may attend reactively during problem periods, and a smaller group may be consistently highly involved.

A negative binomial model would be the natural next step because it generalizes Poisson by adding a dispersion parameter. That extra parameter allows the conditional variance to exceed the conditional mean, which better matches the empirical behavior seen in FSFREQ. In a revised modeling pipeline, the team could keep the same predictor set and recipe structure but swap the outcome model from Poisson to negative binomial, then compare fit using out-of-sample RMSE, deviance-related criteria, and residual diagnostics.

- Why negative binomial is preferable here: it preserves the count-data interpretation while relaxing the unrealistic equal-mean-variance assumption.

- What would improve: standard errors should be more realistic, predicted uncertainty wider, and fitted mean-variance behavior more aligned with the observed count distribution.
- What still would not change: the model would remain associative and observational, so a better count model would improve fit, not establish causality.

Which Models Stood Out, and Why

Two models stood out for different reasons. The lasso logistic model is the preferred binary classifier because it slightly outperformed standard logistic regression while also producing a more compact, easier-to-prioritize set of predictors. The multinomial regression model is the preferred full-outcome model because it preserves the original four-grade structure and therefore communicates more faithfully what the response variable measured.

This split decision is important for communication. If the goal is forecasting who is likely to be a top-performing student, lasso logistic is the sharper tool. If the goal is describing the full distribution of academic outcomes across families, the multinomial model is the better reporting model.

Which Features Best Explain the Research Question

Taken together, the exploratory analysis and the fitted models point to a consistent substantive pattern. Home-based engagement appears to matter more, or at least more consistently, than school-based participation once the full set of predictors is considered.

- Homework frequency and homework-support variables showed the clearest positive association with stronger grades in the exploratory charts and remained relevant in the modeling stage.
- Broad school-involvement measures did not show a simple positive pattern and may sometimes reflect reactive involvement when students are already struggling.
- Student-experience variables, especially school enjoyment and absenteeism, were highly influential and should not be ignored when interpreting family effects.
- Demographic and household background variables remained important, which means family involvement operates within a broader social context rather than on its own.

The cleanest pattern the project can reveal is therefore not “more involvement everywhere is always better.” It is more specific: routine, home-based academic support seems to be the form of involvement most clearly aligned with stronger student performance in this analytic sample.

Limitations and Cautions for Interpretation

- The analysis is observational and cross-sectional, so it identifies associations rather than causal effects.
- Complete-case analysis simplified modeling but may introduce bias if missingness is systematic.
- Class imbalance made the lower-performing grade categories harder to predict in the multinomial model.
- Some coefficient directions depend on factor coding and reference groups, so the most reliable conclusions come from triangulating visuals, model summaries, and comparative performance.

Advice for the GVSU K-12 Team

For GVSU's K-12 Connect, the report suggests that the most actionable opportunity is not simply getting more adults into school buildings. Instead, the stronger recommendation is to help families build sustainable daily learning routines at home.

- Prioritize programs that support homework routines, assignment checking, and practical at home academic monitoring.
- Treat school-event participation as complementary support rather than the main signal of academic success.
- Combine family-engagement initiatives with student-wellbeing supports, especially those connected to school enjoyment and attendance.

In short, the project supports a targeted message: family involvement matters, but the form of involvement matters greatly. The clearest evidence in this analysis favors consistent home-based engagement over school-event participation as a predictor of stronger academic performance.

> "Our project suggests that the most reliable signal of academic success is not just whether families show up at school events, but whether they are able to create steady learning support at home."